

PROBLEM STATEMENT NO 5- INTERACTIVE DASHBOARD & BUSINESS RECOMMENDATIONS

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Course: MBA (Finance & Business Analytics)

Dataset: Sample Dataset (Kaggle)

Mapped CO5: Integrate the Visualizations into an interactive dashboard and recommend business actions

1. Introduction:

This report combines the ten visualizations created in Problem Statement No. 04 into one interactive dashboard using the cleaned Superstore Sales Dataset. The dashboard brings important information about sales, profit, discounts, and regional performance into a single view.

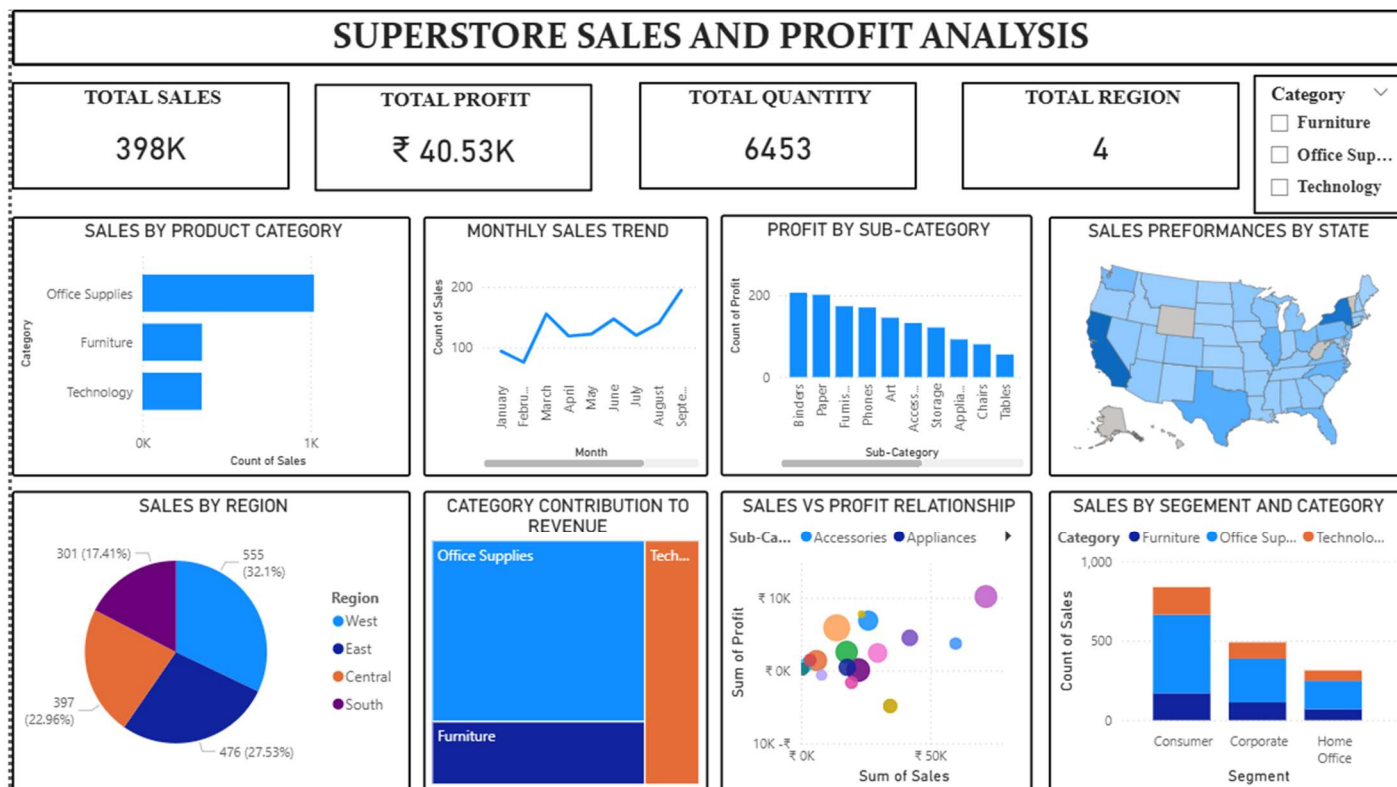
It helps management check important performance indicators quickly and also allows them to explore the data by category, region, segment, and time period. This report explains the dashboard structure, the KPIs used, the main business insights obtained from the visualizations, and the recommendations that can be taken based on these findings.

2. Dashboard Overview:

The interactive dashboard contains the following visualizations on a single dashboard. Slicers and filters are provided for Region, Category, Segment, and Order Date, which allow users to explore the data in more detail.

- Sales by Product Category – Bar Chart
- Profit by Sub-Category – Column Chart
- Monthly Sales Trend – Line Chart
- Sales Contribution by Region – Pie/Donut Chart
- Sales vs. Profit Relationship – Scatter Plot
- Distribution of Order Discounts – Histogram
- Sales, Profit & Quantity by Sub-Category – Bubble Chart
- Category Contribution to Revenue – Tree Map
- Sales Performance by State – Filled Map
- Sales by Segment and Category – Stacked Bar/Column Chart

A screenshot of the completed interactive dashboard created using the selected BI tool, such as Power BI, Tableau, or Excel, should be inserted in this section.



3. List of KPIs Used:

The following KPIs are used in the dashboard to monitor business performance:

KPI| Description

Total Sales| Shows the total revenue generated from all orders.

Total Profit| Shows the overall profit earned from all orders.

Profit Margin (%)| Shows profit as a percentage of sales for different categories and sub-categories.

Average Discount (%)| Shows the average discount given across orders.

Sales by Region/State| Shows how revenue is distributed across different geographic areas.

Sales by Category/Sub-Category| Shows sales and profit performance across different product groups.

Sales by Customer Segment| Shows the contribution of Consumer, Corporate, and Home Office segments to total sales.

Order Quantity| Shows the total number of units sold overall and by product line.

Average Order Value| Shows total sales divided by the number of orders.

Order Processing Time| Shows the average number of days between the order date and shipping date.

4. Business Insights:

Based on the dashboard visualizations, the following business insights can be identified:

1. Uneven Category Performance

Sales performance is different across product categories. Technology and Office Supplies generally contribute higher sales volumes. However, some Furniture sub-categories, such as Tables and Bookcases, have comparatively lower profit margins even though they generate steady sales.

2. Discounts Affect Profitability

Higher discount rates are associated with lower profits. In some cases, orders with high discounts even result in losses. This suggests that the discounting policy needs to consider profitability rather than focusing only on increasing sales.

3. Sales Are Concentrated in Certain Locations

A relatively small number of states and regions contribute a large portion of the total revenue. At the same time, some states have very low sales activity. These areas may represent markets where the business has not yet developed strongly.

4. Seasonal Sales Pattern

The monthly sales trend shows seasonal changes, with sales generally increasing toward the end of the year, especially during Q4. This indicates a seasonal buying pattern that can be considered while planning inventory and marketing activities.

5. Customer Segment Performance

The Corporate and Consumer segments contribute the largest share of sales in most categories. The Home Office segment contributes less in comparison, which indicates that there may be an opportunity to increase sales from this customer group.

6. High Sales Do Not Always Mean High Profit

The Sales vs. Profit scatter plot shows that some orders with high sales have low or even negative profit. This indicates that certain products or discount combinations may generate strong revenue but perform poorly in terms of profit margin.

5. Managerial Recommendations

Based on the insights identified from the dashboard, the following recommendations can be considered:

1. Control Discounts

The company can introduce discount limits or different approval levels for products and sub-categories that are sensitive to discounts. This can help prevent excessive discounting from reducing profit margins.

2. Focus on Underperforming Regions

Marketing and distribution investment can be increased in states and regions where sales are currently low but there is potential for growth. At the same time, existing investment in high-performing regions should be maintained to support continued revenue generation.

3. Review Furniture Pricing and Costs

The company can review the pricing and cost structure of low-margin Furniture sub-categories such as Tables and Bookcases. Possible actions include negotiating better terms with suppliers or making selective price adjustments.

4. Prepare for Q4 Demand

Inventory planning and promotional campaigns should be aligned with the observed Q4 sales increase. Preparing stock and marketing activities in advance can help the company meet seasonal demand and improve order fulfillment.

5. Target the Home Office Segment

Since the Home Office segment contributes less sales compared with Consumer and Corporate customers, the company can develop targeted marketing campaigns specifically for this segment to increase its contribution.

6. Conduct Regular Dashboard Reviews

Management can conduct a regular dashboard review, such as once every month. By monitoring the selected KPIs regularly, management can identify performance changes and respond to emerging issues at an early stage.

6. Conclusion:

The integration of the ten visualizations into one interactive dashboard provides a consolidated view of the Superstore business. It allows management to examine sales, profitability, regional performance, and customer segments through different filters and drill-down options.

The dashboard highlights areas of strong performance, such as high-performing categories and regions, as well as areas that require attention, including excessive discounts, low-margin sub-categories, and lower-performing customer segments.

The recommendations provided in this report connect the dashboard findings with practical business actions. Therefore, the interactive dashboard supports data-based decision-making and fulfills CO5 by connecting data visualization with business recommendations.